AAAI SPRING SYMPOSIUM SERIES (SSS 2026)

Evaluating the Architectural Reasoning Capabilities of LLM Provers via the Obfuscated Natural Number Game

Lixing Li · St Anne's College, Oxford

TL;DR Rename every identifier in the Lean 4 Natural Number Game and the logic stays the same, but the semantic cues are gone. Every model tested slows down significantly; reasoning models keep their accuracy, while general models (GPT-4o, Claude-Sonnet-4.5) lose it.

Proceedings of the AAAI Symposium Series 8(1), 588–591 · Code `github.com/lllx125/Obfuscated-NNG`

 doi.org/10.1609/aaaiss.v8i1.42591

MOTIVATION

Are LLM provers reasoning, or retrieving?

1 THE QUESTION

- LLM provers such as DeepSeek-Prover-V2 do well on **MiniF2F** and **PutnamBench**.
- But modern benchmarks mostly reward **library retrieval**: finding the right high-level lemma.
- In code generation, semantics-preserving perturbations such as identifier renaming measurably degrade performance on LiveCodeBench.

How large is this semantic-reasoning gap in **formal deductive logic**?

2 ARCHITECTURAL REASONING

Synthesizing formal proofs using **only local definitions, axioms, theorems and limited tactics** within a closed theory. It is what AI needs to discover theorems in unexplored, "alien" domains. Three exclusions:

- **No semantic knowledge.** Names like `add` or `succ` cannot be used to infer properties.
- **No external libraries** such as Mathlib.
- **No high-level tactics** (e.g. `linarith`, `simp`); only fundamentals like `rw` and `induction`.

Goal: quantify the **performance tax** of removing semantic identifiers by benchmarking LLMs on the O-NNG and the NNG.

3 WHY THE NATURAL NUMBER GAME (NNG)

- **Closed theory:** every theorem is derived from the ground up using Peano axioms, with no Mathlib.
- **Fits in context:** definitions and earlier theorems are concise, so the model sees all previous knowledge.
- **Restricted tactics:** designed to be solved with fundamental tactics, not automation such as `ring`.

Obfuscating its names gives a **zero-knowledge, closed environment**.

doi.org/10.1609/aaaiss.v8i1.42591

Same logic, alien names: the Obfuscated NNG

1 OBFUSCATION ALGORITHM

$P = \lambda^{2.5}$ · substitute: P · insert: 0.4P · delete: 0.3P

- Noise level $\lambda \in [0, 1]$ sets the per-character perturbation probability P; the exponent samples low noise more densely.
- nlpaug `RandomCharAug` perturbs **every definition, operator, axiom and theorem** name, using ASCII and Lean-allowed Unicode; names stay non-empty.

THEOREM `add_comm` AT EACH NOISE LEVEL (examples from the public repo)

λ = 0	`add_comm (a b : MyNat) : add a b = add b a`
λ = 0.2	`add_KmFm (a b : Mmyat) : add a b = add b a`
λ = 0.4	`add_Φom7 (a b : MθNae) : add a b = add b a`
λ = 0.6	`jad_Qη℃m (a b : oEyab) : Eξ a b = Eξ b a`
λ = 0.8	`μ4dA_c2o (a b : dBNaT) : Ydd a b = Ydd b a`
λ = 1.0	`KΩΦzɑuωT (a b : H0Ja) : KxN a b = KxN b a`

2 BEFORE AND AFTER: A FULL LEAN 4 PROOF

λ = 0 · original NNG — from the public repo

```
theorem add_comm (a b : MyNat) :
    add a b = add b a := by
  induction b with
  | zero => rw [add_zero, zero_add]
  | succ d ih => rw [add_succ, ih, succ_add]
```

λ = 1.0 · fully randomized — from the public repo

```
theorem KΩΦzɑuωT (a b : H0Ja) :
    KxN a b = KxN b a := by
  induction b with
  | GnAθ => rw [E6ZMF_Zζ, ωIξaιvη7]
  | θ7Kr d ih => rw [Qγx1dtΨo, ih, gXqXΔℝvO]
```

Proof structure is untouched; only the names change. The model must rely on the logical structure alone.

doi.org/10.1609/aaaiss.v8i1.42591

EVALUATION SETUP

Five models, 68 problems, six noise levels, checked by Lean

1 SIX-STAGE PIPELINE

1. **Parsing**: extract the NNG in strict dependency order.
2. **Obfuscation**: one dataset per noise level λ.
3. **Query generation**: "alien" definitions, target theorem, previous theorems, available tactics.
4. **Benchmarking**: query the LLMs for Lean 4 proofs.
5. **Verification**: Lean 4 compiler (`v4.27.0`) checks syntax and logic.
6. **Analysis**: success rates, timing, proof plan length.

2 DATASET AND MODELS

68
formal problems from the NNG

8
modules

Addition, Implication, Algorithm, Multiplication, Power, Advanced Addition, Less-Than-or-Equal, Advanced Multiplication

REASONING MODELS

DeepSeek-R1 GPT-5

MATH-SPECIFIC, ALSO REASONING

DeepSeek-Prover-V2

GENERAL-PURPOSE MODELS

GPT-4o Claude-Sonnet-4.5

EXCLUDED (API-ONLY EVALUATION)

~~Goedel-Prover-v2~~

3 PROTOCOL AND METRICS

- Models return JSON: a natural-language **"Draft"** proof plan and a **"Code"** Lean 4 block passed straight to the compiler.
- **6** noise levels: $\lambda \in \{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$; $\lambda = 0$ is the original NNG.
- Each model queried **five times per theorem** at every noise level.
- **Correct Rate (%)**: correct problems out of 68.
- **Average Time (s)**: from sending the query to receiving the response.
- Significance: **one-way ANOVA** across noise levels.

doi.org/10.1609/aaaiss.v8i1.42591

RESULTS

Accuracy and solving time as the noise level λ rises

Top row: Correct Rate (%) **Bottom row**: Average Time (s) λ = 0 is the original NNG; λ = 1.0 is fully randomized identifiers

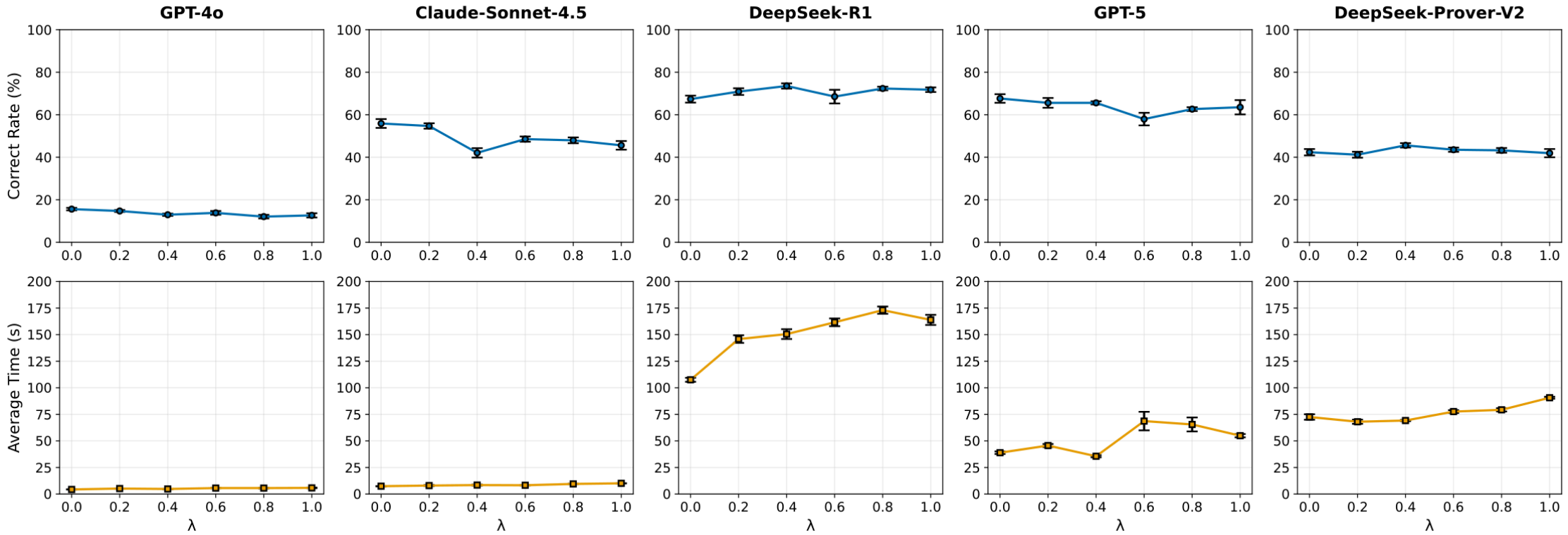


Figure 1 of the paper. Error bars represent standard deviation over 5 independent runs.

doi.org/10.1609/aaaiss.v8i1.42591

A universal latency tax, and a split in robustness

1 ONE-WAY ANOVA P-VALUES (TABLE 1)

MODEL	CORRECT RATE	AVERAGE TIME
GENERAL-PURPOSE		
GPT-4o	0.0242*	0.0004*
Claude-Sonnet-4.5	0.0001*	0.0000*
REASONING		
DeepSeek-R1	0.1573	0.0000*
GPT-5	0.0863	0.0001*
DeepSeek-Prover-V2	0.3077	0.0000*

* Indicates statistical significance ($p < 0.05$). Rows regrouped by model type; values as printed in Table 1.

2 WHAT IT SHOWS

5 / 5

models significantly slower on the O-NNG than the NNG. Removing names appears to force a more intensive search or reconstruction of the logical context.

3 / 5

reasoning models kept their accuracy: no significant drop for DeepSeek-R1, GPT-5 or DeepSeek-Prover-V2.

2 / 5

general models lost accuracy for $\lambda > 0$, suggesting reliance on semantic pattern matching.

doi.org/10.1609/aaaiss.v8i1.42591

DISCUSSION

Takeaways, limitations and next steps

1 TAKEAWAYS

- O-NNG gives a **quantitative metric** for the semantic-reasoning gap: a template for zero-knowledge test environments.
- The divergence supports prioritizing **reasoning-focused architectures** for AI4Math.
- Architectural Reasoning is **not computationally free**: the latency tax suggests an intensive internal search.

Relevant to knowledge-grounded semantic agents working with incomplete or obscured semantics.

2 LIMITATIONS

- Renaming keeps the **structure of Peano arithmetic**, so a model could in principle map it back to known math. In the experiment, no LLM attempted to map terms back or referenced Peano theory.
- A stricter test needs **entirely novel axiomatic domains**.
- Five frontier models and 68 natural-number problems; results may not transfer to mid-sized open models or to algebra or topology.

The paper frames this study as a **preliminary exploration**.

3 NEXT STEPS

- A possible **"Reasoning Uncanny Valley"**: partial noise occasionally hurts more than total obfuscation, as residual cues distract.
- Test whether more noise consistently reduces hallucination from familiar names.
- Evaluate **interactively** with real-time Lean error feedback, to see whether self-correction reduces the latency tax.

Longer term: synthesize entirely novel, non-standard axiomatic domains.

doi.org/10.1609/aaaiss.v8i1.42591